
1. Executive Summary & Objective

Project Starlight delivers Pluto's next-generation offline meeting assistant capabilities. The objective is to provide an executive-level second brain that captures, transcribes, and synthesizes conversations with zero cloud leakage, operating entirely on-device with sub-3-second note synthesis on Apple Silicon hardware.

In this phase, Pluto introduces **Reference Document Ingestion**, enabling users to attach project briefs, technical specs, and meeting agendas directly to specific meetings to ground and enrich LLM notes generation with exact domain context.

2. Team & Key Stakeholders

Name	Role	Core Responsibilities
Maya Lin	Staff Platform Architect	SQLite persistence, junction schema design, vector indexing, and revision hashing.
Alex Chen	Lead Product Designer	Editorial meeting notes UX, zero-footprint attachment bar, and document inspector.
David Rossi	Native Audio Lead	CoreAudio ring-buffer synchronization, device switching resilience, and echo cancellation.
Priya Sharma	Machine Learning Lead	Whisper.cpp CoreML optimization, speaker diarization, and voiceprint clustering.

3. Architecture & Technical Constraints

- ### 3.1 Local Artifacts Junction Pipeline
- Database Schema**: Reference documents are linked to meetings via the `meeting_local_artifacts` junction table (`meeting_id` ↔ `artifact_id`) with cascading deletes on meeting or document removal.
 - Cache Invalidation & Revision Hashes**: Attaching or detaching an artifact computes a SHA-256 hash across both user notes and attached document text (`userNotesHash`). Any modification invalidates outdated cached synthesis runs, prompting the user to regenerate notes.
 - Context Window Budget**: To prevent prompt saturation on 8k/16k context local models, extracted text injected into the LLM synthesis prompt is capped at 48KB per meeting, prioritised by relevance and recency.

- ### 3.2 Supported File Formats & Extraction
- Microsoft Word (.docx)**: Unzipped XML stream parser via JSZip extracting body paragraphs and tables.
 - Apple Pages (.pages)**: Native zip extraction inspecting protobuf index data and quick-look previews.
 - Portable Document Format (.pdf)**: Extracted via `unpdf` with page boundary normalisation and whitespace cleanup.
 - Plaintext & Markdown (.md, .txt)**: UTF-8 normalized stream ingestion.

- ### 3.3 Privacy & Grounding Guarantees
- Raw audio, transcripts, and attached document text must never leave the user's machine without explicit export actions.

- Notes synthesis must explicitly cite and ground claims in evidence: statements from attached documents are distinguished from spoken dialogue using `[Attached Document: <Title>]` markers.

4. Q3 Sprint Milestones & Deliverables

Milestone 1: Document Ingestion & Impeccable Notes UI

- ****Target Completion****: October 3, 2026
- ****Owners****: Alex Chen & Maya Lin
- ****Success Criteria****:
 - Zero-footprint empty state when no documents are attached.
 - Document attachment accessible via dropdown (`...` menu) and instant drag-and-drop on the notes surface.
 - Quiet inline badge row (`